

472 ✓
274 ✓
427 ✓

Module 4

Chapter 1

Introduction to Combinatorics

4 7 2
↙ ↘
order Matters order doesn't matter

What is combinatorics?

Simply put, combinatorics is “*the art of counting*”! Any problem that needs the answer to “how many” can be addressed by the knowledge of combinatorics.

In this chapter we will learn two basic ways of counting/enumerating outcomes, ***Permutations and Combinations***, which will presage our understanding of probability which we will learn in the next chapter.

Think of the two sentences:

“I need pineapples, chicken and jalapenos as my pizza toppings!”

“The key to my safe is 4-7-2”

For the first sentence, does it matter in which order we place our toppings on the pizza?

How about the second one? Does the order of the digits matter?

This tells us that there are cases where the order in which we enumerate things matter and there are other cases where the order doesn't matter.

In mathematics, when ***order doesn't matter***, it's a ***combination*** and when ***order does matter***, it's a ***permutation***.

Permutations

Let's say we have 8 people:

Alice, Bob, Charles, Don, Evan, Frank, George and Hannah

In how many ways can we award three prizes (Gold, Silver and Bronze) among these 8 participants?

A B C D E F G H 8 choices

B C D E F G H 7 choices

C D E F G H 6 choices

In this case, the order in which we give the prizes matters. Hence, we will be using Permutations here:

Let's look at our options for the *Gold medal*:

We have **8 options**: A, B, C, D, E, F, G, H. Let's say A wins the Gold Medal.

This means we have **7 options** left for the *silver medal*: B, C, D, E, F, G, H. Let's assume B wins the Silver Medal.

Now, for the *bronze medal*, we have **6 options** left: C, D, E, F, G, H. let's say C wins the Bronze!

We picked certain people who won, but that doesn't really matter. We can make other choices and we will still have the same number of options: 8 for gold, 7 for silver and 6 for bronze: **8*7*6 total choices!**

Here's what we did:

We had to order 3 people out of 8. To do this, we started with all options (8) then took them away one at a time (7, then 6) until we ran out of medals.

We know that the factorial of 8 is given by:

$$8! = 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1$$

But we don't need so many options. We can see that we need to **stop at 5!**. How can we do that?

We, essentially, need to **get rid of 5!** from the formula for 8!

So, if we do $\frac{8!}{5!}$, we get:

permutation

$$\frac{n!}{(n-k)!}$$

$$\frac{8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1}{5 \times 4 \times 3 \times 2 \times 1} = 8 \times 7 \times 6 = 336$$

$$3 \times 2 \times 1$$

A B C

$$3!$$

How did we get the 5? Well, that was what we were left with after we picked 3 medals from 8. So, a better way of writing the above formula would be:

$$\frac{8!}{(8 - 3)!}$$

K

In more general terms, if we have ***n options*** and we want to ***pick k*** out of those, in a ***certain order***, we can get the total number of ways of doing so by using the ***Permutation formula***:

$$P(n, k) = \frac{n!}{(n - k)!}$$

Combinations

Combinations are a little different. Let's say I still have the same contestants but now, instead of giving them a separate gold, silver and bronze medal, I decide to give them ***three prizes of the same value***.

In how many ways can I give 3 prizes to 8 people?

In this case, the ***order doesn't matter***. If I give the prize to Alice, Bob and Charles, it's the same as giving it to Bob, Alice and Charles. It seems like using Permutations will lead to redundancies here.

Let's understand how to calculate the ***redundancies*** here:

We can say, Alice, Bob, Charlie = Charlie, Alice, Bob. How many such redundancies do we have? Well, it's ***the total number of ways we can arrange 3 people***:

We can think of it as "Number of ways we can put three people in three empty spaces":

We have 3 options for the first space, 2 for the second and 1 for the last space:
 $3 \times 2 \times 1 = 3! = 6$

In general, if you have ***n*** people and you want to know ***how many ways you can arrange them***, it is simply ***n!***

$$\frac{n!}{(n-k)!k!}$$

So, in this case, for every choice we pick, we have $3!$ variations for each: For e.g., if our choice is giving the prizes to Alice, Bob and Charlie, then there are 6 variations of it: $ABC=ACB=BCA=CBA=BAC=CAB$

So, if we want to figure out how many combinations we have, we just calculate the permutations and ***get rid of the redundancies***. In this case, we had 336

permutations and we divide it by $3!$ for each permutation and that gives us $\frac{336}{6} = \frac{8!}{(8-3)! \cdot 3!}$

56

This gives us the general ***formula for combination***:

$$C(n, k) = \frac{n!}{(n - k)! (k)!}$$

Let's Practice

1. A student has to answer 10 questions, choosing atleast 4 questions from part A and at least 4 from part B. There are 6 questions in part A and 7 in part B. In how many different ways can they answer 10 questions?
2. Find the number of ways in which the letters of the word INDEPENDENCE can be arranged. In how many of these arrangements
 - Do the words start with P
 - Do the vowels all come together?
 - Do the vowels never occur come all together
 - Do the words begin with I and end with P
3. There are 10 lamps in a hall and each of them can be switched on independently. In how many different ways can the hall be illuminated?
4. Determine the number of 5 card combinations in a deck of 52 cards if there is exactly one Ace in each combination.
5. You are babysitting for kids: Lancer, Prancer, Gloopin, Bloopin! You want to arrange them in a line for a game you want to play with them. However Lancer and Plancer do not get along and hence cannot be placed next to each other. In how many different ways can you arrange these kids in a line without Lancer and Plancer fighting?

1. Total Questions to be answered = 10

A → min 4 questions

Total → 6 Questions

B → min 4 questions

Total → 7 Questions

order doesn't matter

Case 1

4 from A

and

6 from B

Two events happening at the same time
↓
Multiplication

$${}^6C_4 \times {}^7C_6 = 105$$

Case 2

5 from A and 5 from B

$${}^6C_5 \times {}^7C_5 = 126$$

Case 3

6 from A and 4 from B

$${}^6C_6 \times {}^7C_4 = 35$$

Total # of ways:

$$105 \oplus 126 + 35 = 266$$

OR → uses $\oplus$

2. INDEPENDENCE

$$I \rightarrow 1$$

$$N \rightarrow 3$$

$$D \rightarrow 2$$

$$E \rightarrow 4$$

$$P \rightarrow 1$$

$$C \rightarrow 1$$

$$\text{Total: } 12$$

of ways in which we can arrange the letters
Order Matters

$$\frac{12!}{3! \cdot 4! \cdot 2!}$$

↓
getting rid of redundancies for repeated letters

words starting with P:

P _ _ _ _ _

11 empty spaces

$$\frac{11!}{3! \cdot 4! \cdot 2!}$$

of words where vowels all come together.

"I" and "E"s come together

Putting I and Es together as 1 letter,
we now have 8 letters

I and Es can also be arranged in

$$\frac{(4+1)!}{4!} \text{ ways} = \frac{5!}{4!} = 5 \text{ ways}$$

∴ Total # of ways:

$$\frac{8!}{3! 2!} \times 5$$

of arrangements where vowels never come together:

of arrangements - # of arrangements where vowels come together:

$$\frac{12!}{3! 4! 2!} - \frac{5 \times 8!}{3! 2!}$$

of arrangements that begin with I and end with P.

I - - - - - P

10 empty spaces

$$\frac{10!}{3! 4! 2!}$$

3. There needs to be at least one lamp "ON" for the room to be illuminated.
- For each lamp, we have 2 options: ON and OFF
- We have 10 lamps

$\therefore$ Total # of combinations

$$2 \times 2 \times 2 \dots 10 \text{ terms} = 2^{10}$$

One of the above combinations is "all lamps off". We need to get rid of that

$\therefore$ Different ways the hall can be illuminated

$$= 2^{10} - 1$$

4. Ace - - - -

Order doesn't matter

of ways we can choose 1 Ace out of

4 Aces: 4C_1

of ways we can choose 4 cards out of remaining 48 cards: ${}^{48}C_4$

∴ Total # of ways: ${}^4C_1 \times {}^{48}C_4$

5. L P Q B

of ways they can be arranged: $4!$

Let L and P be together.

(LP) Q B : # of ways L and P are together in the arrangement:

$3! \times 2!$

of ways L and P can be arranged among themselves

∴ # of arrangements where L and P aren't together

$4! - 3! \times 2! = 12 \text{ ways}$